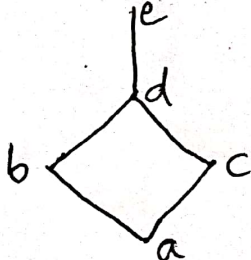


Module -2			
Q. No.	Questions	Bloom's LL	COs
1	Find Explicit form for the recursive relation $a_1 = 4$, $a_n = a_{n-1} + n$, $n \geq 2$	L3	CO2
2	If $F_0, F_1, \dots$ are Fibonacci numbers then Prove that $\sum_{i=1}^n \frac{F_{i-1}}{2^i} = 1 - \frac{F_{n+2}}{2^n}$	L2	CO2
3	How many arrangement are there for all the letters in the word "SOCIOLOGICAL". In how many of these arrangements i) A and G are Adjacent ii) All Vowels are Adjacent.	L3	CO2
4	How many positive integers n to form using the digits 3, 4, 4, 5, 5, 6, 7. If we want n to exceed 5,000, 000 ?	L3	CO2
5	If all permutations of the letters of the word "AGAIN" are arranged in the order as in dictionary. what is the 49 th word ?	L3	CO2
6	A student as to answer 10 questions, choosing at least 4 from each of part A and part B. If there are 6 questions in part A and 7 in part B, in how many ways can the student choose 10 questions?	L3	CO2
7	A women as 11 closed relatives and she wishes to invite 5 of them to dinner. In how many ways can she invite them in the following situations i) There is no restriction on the choice ii) 2 particular persons will not attend separately iii) 2 particular persons will not attend together	L3	CO2
8	Find the Co-efficient of i) $x^9 y^3$ in the expansion of $(2x - 3y)^{12}$ ii) x^{12} in the expansion of $x^3(1 - 2x)^{10}$	L3	CO2
9	Determine Co-efficient of x^6 in the expansion of $(3x^2 - (2/x))^{15}$	L3	CO2
Module -3			
1	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$. If $(g \circ f)(x) = 9x^2 - 9x + 3$. Determine a and b.	L3	CO3
2	If $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6\}$ i) How many functions are there from A to B? How many of these are one-to-one? How many are onto? ii) How many functions are there from A to B? How many of these are one-to-one? How many are onto?	L3	CO3

3	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 3x - 5, & x > 0 \\ 1 - 3x, & x \leq 0 \end{cases}$ Find $f(0)$, $f(-1)$, $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(3)$, $f^{-1}([-5, 5])$	L3	CO3
4	If $A = \{1, 2, 3, 4, 5\} \times \{1, 2, 3, 4, 5, 6\}$ and define R on A by $(x_1, y_1)R(x_2, y_2)$ if $x_1 + y_1 = x_2 + y_2$ i) Verify that R is an equivalence relation on A ii) Determine the equivalence classes $[(1, 3)]$, $[(2, 4)]$ and $[(1, 1)]$	L3	CO3
5	Let $A = B = C = \mathbb{R}$ and $f: A \rightarrow B$ and $g: B \rightarrow C$ be defined by $f(a) = 2a + 1$, $g(b) = \frac{1}{3}b$, $\forall a \in A, \forall b \in B$. Compute $(g \circ f)$ and show that $(g \circ f)$ is invertible. What is $(g \circ f)^{-1}$?	L3	CO3
6	If R is an Equivalence relation on set A and $x, y \in A$ then Prove that i) $x = [x]$ ii) xRy iff $[x] = [y]$ and iii) If $[x] \cap [y] = \emptyset$ then $[x] = [y]$.	L2	CO3
7	Define an Equivalence relation? Draw the Hasse diagram representing the positive divisors of 36.	L3	CO3
8	For $A = \{a, b, c, d, e\}$ the Hasse diagram for the poset (A, R) is shown in Fig. i) Determine the relation matrix for R ii) Construct the directed graph G that is associated with R . 	L3	CO3
9	State Pigeonhole principle. Let ABC is an equilateral triangle whose sides are of length 1cm each. If we select 5 points inside the triangle. Prove that atleast two of their points are such that the distance between them is less than $\frac{1}{2}$ cm.	L2	CO3
10	If $A = \{1, 2, 3, 4\}$, R and S are relations on A defined by $R = \{(1, 2), (1, 3), (2, 4), (4, 4)\}$, $S = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4)\}$. Find $R \circ S$, $S \circ R$, R^2 , S^2 and write down their matrices.	L3	CO3
11	Let $A = \{1, 2, 3, 4\}$ and R be a relation on A defined by xRy iff x divides y . Then i) Write R as ordered pairs ii) Draw Diagram iii) Write matrix of R .	L3	CO3

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